# Systems of First Order Linear ODEs

## 1 First order ODE systems

The general system of first order equations involving $n$ dependent variables defined on an interval $I \subseteq \mathbb{R}$ can be written as

$$\begin{cases} y_1'(t) = F_1(t, y_1, \dots, y_n) \\ y_2'(t) = F_2(t, y_1, \dots, y_n) \\ \quad\vdots \\ y_n'(t) = F_n(t, y_1, \dots, y_n) \end{cases} \qquad \text{for} \quad t \in I. \tag{1}$$

$$\text{with initial conditions} \quad y_1(t_0) = x_1, \quad y_2(t_0) = x_2, \quad \dots, \quad y_n(t_0) = x_n,$$

The solution to the system can be expressed in vector form as

$$\boldsymbol{y}(t) = \begin{pmatrix} y_1(t) \\ y_2(t) \\ \vdots \\ y_n(t) \end{pmatrix}$$

**Theorem 1.1** (Existence and Uniqueness).

For the system of first order ODEs (1), if the functions $F_1, F_2, \dots, F_n$ and the partial derivatives $\dfrac{\partial F_i}{\partial y_j}$ for all pairs $i, j = 1, 2, \dots, n$ are continuous on a region $R$ defined by

$$R := [t_0 - a, t_0 + a] \times [x_1 - b, x_1 + b] \times \cdots \times [x_n - b, x_n + b] \subseteq \mathbb{R}^{n+1},$$

where constants $a, b > 0$, then there exists a unique solution $\boldsymbol{y}(t)$ for $t \in (t_0 - h, t_0 + h)$ to the IVP, where

$$h = \min\{a, b/M\} \quad \text{and} \quad M = \max_{\boldsymbol{v} \in R}\{|F_1(\boldsymbol{v})|, |F_2(\boldsymbol{v})|, \dots, |F_n(\boldsymbol{v})|\}.$$

Any $n$-th order ODE can be rewritten as a system of $n$ first order ODEs. Given an $n$-th order ODE

$$y^{(n)} = F(t, y, y', \dots, y^{(n-1)}),$$

setting $z_1 = y, z_2 = y', \dots, z_n = y^{(n-1)}$, we obtain the system

$$\begin{cases} z_1' = z_2 \\ z_2' = z_3 \\ \quad\vdots \\ z_{n-1}' = z_n \\ z_n' = F(t, z_1, z_2, \dots, z_n) \end{cases}$$

which is a system of $n$ first order ODEs.

**Definition 1.1** (First-Order Linear System).

A system of first order ODEs is called **linear** if it can be expressed in the form

$$\boldsymbol{y}'(t) = \boldsymbol{P}(t)\boldsymbol{y}(t) + \boldsymbol{g}(t) \qquad \text{for} \quad t \in I, \tag{2}$$

where

$$\boldsymbol{y}(t) = \begin{pmatrix} y_1(t) \\ y_2(t) \\ \vdots \\ y_n(t) \end{pmatrix} \qquad \boldsymbol{g}(t) = \begin{pmatrix} g_1(t) \\ g_2(t) \\ \vdots \\ g_n(t) \end{pmatrix} \qquad \boldsymbol{P}(t) = \begin{pmatrix} p_{11}(t) & p_{12}(t) & \cdots & p_{1n}(t) \\ p_{21}(t) & p_{22}(t) & \cdots & p_{2n}(t) \\ \vdots & \vdots & \ddots & \vdots \\ p_{n1}(t) & p_{n2}(t) & \cdots & p_{nn}(t) \end{pmatrix}$$

**Definition 1.2** (Homogeneous First-Order Linear System).

A first order linear system is called **homogeneous** if $\boldsymbol{g}(t) = \boldsymbol{0}_n$ for all $t \in I$, i.e., it can be expressed as

$$\boldsymbol{y}'(t) = \boldsymbol{P}(t)\boldsymbol{y}(t) \qquad \text{for} \quad t \in I.$$

**Theorem 1.2** (Existence and Uniqueness for First-Order Linear Systems).

For the first-order linear system (2), if all entries of $\boldsymbol{P}(t)$ and $\boldsymbol{g}(t)$ are continuous on an interval $I$, then for any $t_0 \in I$ and $\boldsymbol{y}_0 \in \mathbb{R}^n$, there exists a unique solution $\boldsymbol{y}(t)$ to the IVP

$$\begin{cases} \boldsymbol{y}'(t) = \boldsymbol{P}(t)\boldsymbol{y}(t) + \boldsymbol{g}(t) \\ \boldsymbol{y}(t_0) = \boldsymbol{a} \in \mathbb{R}^n \end{cases} \qquad \text{for} \quad t \in I.$$

In the rest parts, we will focus on the following homogeneous system defined on an open interval $I \subseteq \mathbb{R}$:

$$\boldsymbol{y}'(t) = \boldsymbol{P}(t)\boldsymbol{y}(t) \quad \text{for} \quad t \in I, \tag{3}$$

where $\boldsymbol{y} \in C^1(I)^n$ and $\boldsymbol{P} \in C(I)^{n\times n}$, i.e., all entries of $\boldsymbol{P}$ and $\boldsymbol{y}$ are continuous on $I$.

**Theorem 1.3** (Principle of Superposition).

If $\boldsymbol{y}_1(t)$ and $\boldsymbol{y}_2(t)$ are solutions to the homogeneous first-order linear system (3), then the linear combination

$$\boldsymbol{y}(t) = c_1\boldsymbol{y}_1(t) + c_2\boldsymbol{y}_2(t)$$

is also a solution for any constants $c_1, c_2$.

**Definition 1.3** (Wronskian)**.**

The **Wronskian** of $n$ vector functions $\boldsymbol{f}_1(t), \boldsymbol{f}_2(t), \dots, \boldsymbol{f}_n(t) \in C(I)^n$ is defined as

$$W(\boldsymbol{f}_1, \boldsymbol{f}_2, \dots, \boldsymbol{f}_n)(t) \quad := \quad \det\left(\begin{bmatrix} | & & | \\ \boldsymbol{f}_1(t) & \cdots & \boldsymbol{f}_n(t) \\ | & & | \end{bmatrix}\right) \quad \text{for} \quad t \in I.$$

*Remark.* The Wronskian for vector functions is different from the Wronskian for scalar functions. There is no derivative involved in the definition of the Wronskian for vector functions.

**Theorem 1.4** (Liouville's Formula)**.**

Let $\boldsymbol{y}^{(1)}(t), \boldsymbol{y}^{(2)}(t), \dots, \boldsymbol{y}^{(n)}(t)$ be $n$ solutions to the homogeneous system (3). Then the Wronskian of these solutions is given by

$$W(\boldsymbol{y}^{(1)}, \boldsymbol{y}^{(2)}, \dots, \boldsymbol{y}^{(n)})(t) = c \cdot \exp\left(\int \operatorname{tr}(\boldsymbol{P}(t))\,\mathrm{d}t\right) \quad \text{for} \quad t \in I,$$

where $c$ is a constant not depending on $t$.

*Remark.* The **trace** of a square matrix $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ is defined as the sum of its diagonal entries:

$$\operatorname{tr}(\boldsymbol{A}) := a_{11} + a_{22} + \cdots + a_{nn}.$$

***Proof*** (Liouville's Formula)**.**

Before the proof, we need to introduce several theorems from linear algebra:

1. For any square matrix $\boldsymbol{A}$,
$$\frac{\mathrm{d}}{\mathrm{d}t}(\det(\boldsymbol{A})) = \operatorname{tr}\left(\operatorname{adj}(\boldsymbol{A})\frac{\mathrm{d}\boldsymbol{A}}{\mathrm{d}t}\right). \tag{4}$$
2. The trace remains unchanged under cycle permutations:
$$\operatorname{tr}(\boldsymbol{ABC}) = \operatorname{tr}(\boldsymbol{BCA}). \tag{5}$$
3. Property of the adjugate matrix:
$$\operatorname{adj}(\boldsymbol{A})\boldsymbol{A} = \boldsymbol{A}\operatorname{adj}(\boldsymbol{A}) = \det(\boldsymbol{A})\boldsymbol{I}. \tag{6}$$

Now, let $\boldsymbol{Y} := \begin{bmatrix} \boldsymbol{y}^{(1)} & \boldsymbol{y}^{(2)} & \cdots & \boldsymbol{y}^{(n)} \end{bmatrix} \in C^1(I)^{n\times n}$ and $W := \det(\boldsymbol{Y})$ be the Wronskian. Then we have

$$\begin{aligned} W' &= \operatorname{tr}(\operatorname{adj}(\boldsymbol{Y})\boldsymbol{Y}') \\ &= \operatorname{tr}(\operatorname{adj}(\boldsymbol{Y})\boldsymbol{PY}) \qquad \text{since} \quad \boldsymbol{y}' = \boldsymbol{Py} \\ &= \operatorname{tr}(\boldsymbol{Y}\operatorname{adj}(\boldsymbol{Y})\boldsymbol{P}) \\ &= \operatorname{tr}(W\boldsymbol{IP}) \\ &= W\operatorname{tr}(\boldsymbol{P}). \end{aligned}$$

Hence the Liouville's formula. □

**Theorem 1.5.**

If $\boldsymbol{y}^{(1)}, \boldsymbol{y}^{(2)}, \dots, \boldsymbol{y}^{(n)}$ are $n$ solutions to a homogeneous system (3), then

$$\forall t \in I, \quad W(\boldsymbol{y}^{(1)}, \boldsymbol{y}^{(2)}, \dots, \boldsymbol{y}^{(n)})(t) = 0 \qquad \Longleftrightarrow \qquad \exists t_0 \in I \text{ s.t. } W(\boldsymbol{y}^{(1)}, \boldsymbol{y}^{(2)}, \dots, \boldsymbol{y}^{(n)})(t_0) = 0.$$

***Proof*.**

Proven by Liouville's formula. □

**Definition 1.4** (Fundamental Set of Solutions)**.**

A set of $n$ **linearly independent** solutions $\{\boldsymbol{y}^{(1)}(t), \boldsymbol{y}^{(2)}(t), \dots, \boldsymbol{y}^{(n)}(t)\}$ to the homogeneous system (3) is called a **fundamental set of solutions**.

**Theorem 1.6.**

If the $n$ solutions $\boldsymbol{y}^{(1)}(t), \boldsymbol{y}^{(2)}(t), \dots, \boldsymbol{y}^{(n)}(t)$ of the homogeneous system (3) are **linearly independent** for each $t \in I$, then every solution of the system can be expressed as a linear combination:

$$\boldsymbol{y}(t) = c_1\boldsymbol{y}^{(1)}(t) + c_2\boldsymbol{y}^{(2)}(t) + \cdots + c_n\boldsymbol{y}^{(n)}(t) \quad \text{for} \quad t \in I$$

in **exactly one way**.

***Proof*.**

The Principle of Superposition indicates the solutions form a vector space, thus there exist bases. Therefore, the essence of this proof is to show that the dimension of the solution space is exactly $n$, so that $n$ linearly independent solutions form a basis (FSS). Also, since the solution space is a subspace of $C^1(I)^n$, the dimension of the solution space cannot exceed $n$. Hence, we only need to show that the dimension is at least $n$.

□